QAOA for MaxCut

The Quantum Approximate Optimization Algorithm

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Physics 565 / 656 · Spring 2026

Introduction

Definition of a graph

Vertex set

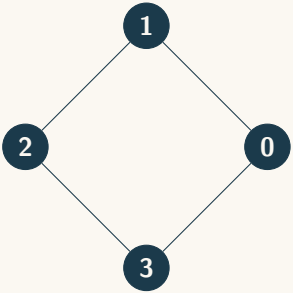
$V = \{0, 1, 2, 3\}, \quad |V| = n = 4$

Edge set

$E \subseteq \binom{V}{2} = \{\{0,1\}, \{1,2\}, \{2,3\}, \{0,3\}\}$

Notation

Throughout: $n = |V|$ qubits, $m = |E|$ edges.
Cuts will be encoded as bitstrings $z \in \{0,1\}^n$.



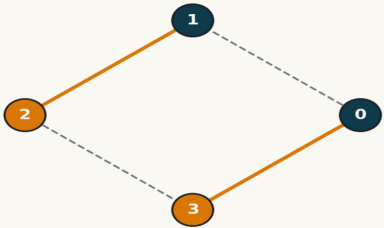
C_4 : cycle on $n = 4$ vertices, $|E| = 4$

The MaxCut problem

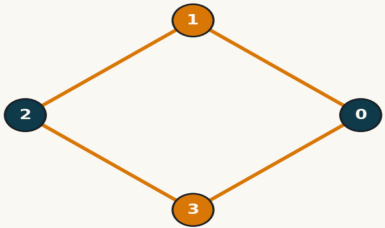
$$C(S) \;=\; \big|\{(i,j) \in E : i \in S,\; j \in \overline{S}\}\big|, \qquad S^\star \in \arg\max_{S \subseteq V} C(S).$$

Partition $V = S \sqcup \overline{S}$ to maximise the number of cut edges

Suboptimal: S = {0, 1} cut = 2



MaxCut: S = {0, 2} cut = 4



For C_4: 2^4 = 16 labelled partitions / 2^3 = 8 distinct cuts. The maximum is 4.

Motivation

A worked example: portfolio diversification (1/2)

The investor's problem.

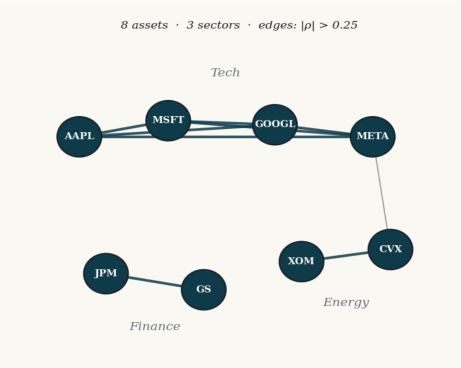
Hold n assets with daily returns $r_i(t)$. *Risk concentrates when assets move together* — so split the portfolio into two halves $S, \overline{S}$, each *internally uncorrelated*, so a shock to one asset does not drag the rest of its half with it.

Measure correlation.

$$\rho_{ij} \;=\; \frac{\mathrm{Cov}(r_i, r_j)}{\sigma_i\,\sigma_j} \;\in\; [-1, 1]$$

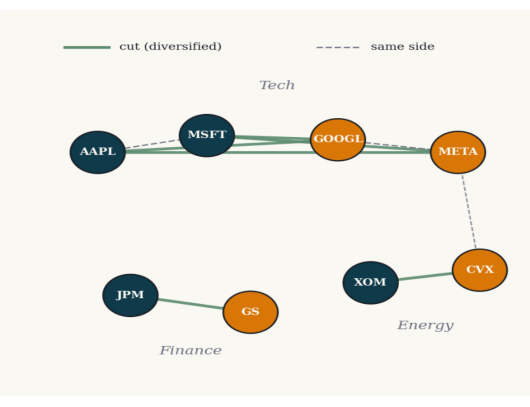
Build the correlation graph.

- Vertex $\leftrightarrow$ asset.
- Edge (i, j) if $|\rho_{ij}| > 0.25$.
- Same-sector pairs $\Rightarrow$ thick edges ($\rho \approx 0.87$).



8 assets, 3 sectors — intra-sector edges dominate.

A worked example: portfolio diversification (2/2)



Cut edges (green) separate correlated pairs; same-side edges (dashed) are correlations within a group.

Result on the 8-asset example.

Group $S = \{\text{AAPL, MSFT, JPM, XOM}\}$,
Group $\overline{S} = \{\text{GOOGL, META, GS, CVX}\}$.
Each group contains one asset from every sector — automatic sector diversification.

Strategy	Avg. $\lvert\rho\rvert$ within group
Naïve (Tech | Fin+Energy)	0.87 / 0.32
MaxCut	**0.22 / 0.26**

$\sim 4\times$ less correlated inside each group.

Stylised: we threshold at $\lvert\rho\rvert > 0.25$ and ignore expected returns and budget constraints — the point is that "diversification" maps onto a cut.

Theory

From bitstring to Hamiltonian

1. Decision variable

$z_i \in \{0,1\}$ with rule $z_i = 0 \Leftrightarrow i \in S$, $z_i = 1 \Leftrightarrow i \in \overline{S}$.

Cut size: $C(z) = \sum_{(i,j)\in E} \mathbf{1}[z_i \neq z_j]$.

2. Rewrite with spins

$s_i = 1 - 2z_i \in \{+1, -1\}$:

$$\mathbf{1}[z_i \neq z_j] \;=\; \tfrac{1}{2}(1 - s_i s_j).$$

3. Quantise $s_i \to Z_i$ (Pauli-Z on qubit i):

$$\boxed{H_C \;=\; \sum_{(i,j)\in E} \tfrac{1}{2}(I - Z_i Z_j)}$$

Key property

H_C is *diagonal* in the computational basis:

$$H_C \left|z\right\rangle = C(z) \left|z\right\rangle.$$

So $\langle H_C \rangle$ is the expected cut value, and the ground states are the MaxCut bitstrings.

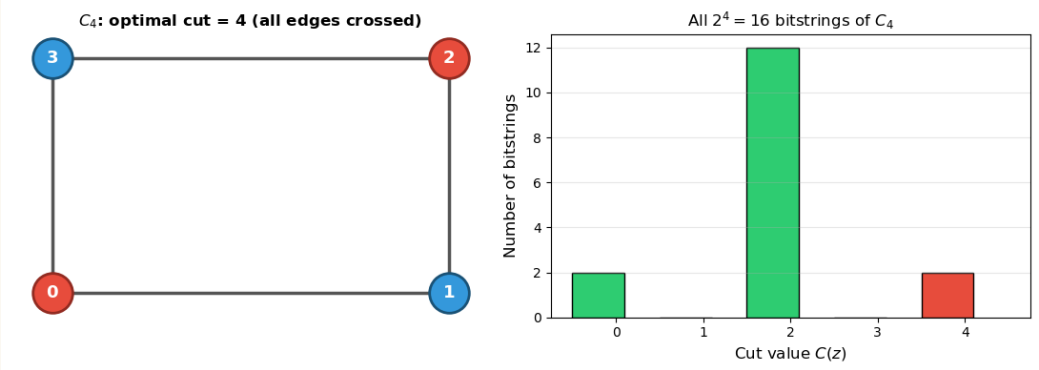
Goal

Prepare $|\psi\rangle$ that maximises $\langle\psi|H_C|\psi\rangle$.

What we are looking for

Measurement rule

Measuring $|\psi\rangle$ in the computational basis returns $z = z_0 z_1 \cdots z_{n-1}$. Read out:
$z_i = 0 \Rightarrow i \in S, \;\; z_i = 1 \Rightarrow i \in \overline{S}$.



Goal: maximise the probability of sampling $|1010\rangle$ or $|0101\rangle$ — the two MaxCut solutions of C_4.

Starting state: $|+\rangle^{\otimes n}$

$$|+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle = H^{\otimes n} |0\rangle^{\otimes n}.$$

1. Uniform over candidates. Equal weight on all 2^n bitstrings — no partition is preferred a priori.

2. Aligned with the mixer. $X|+\rangle = +|+\rangle$, so $H_B = \sum_k X_k$ has $|+\rangle^{\otimes n}$ as its top eigenstate.

3. Cheap to prepare. One Hadamard per qubit — constant-depth circuit from $|0\rangle^{\otimes n}$.

4. Adiabatic starting point. Ground state of $H(t=0) = H_B$ in the adiabatic interpolation

$$H(t) = (1 - t/T)H_B + (t/T)H_C.$$

The QAOA ansatz

State

$$|\psi_p(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle \;=\; U_B(\beta_p)U_C(\gamma_p)\cdots U_B(\beta_1)U_C(\gamma_1)\;|+\rangle^{\otimes n}$$

Cost unitary

$$U_C(\gamma) = e^{-i\gamma H_C} \;=\; \prod_{(i,j)\in E} e^{-i\gamma(I - Z_i Z_j)/2}$$

Encodes problem structure — entangling.

Mixer unitary

$$U_B(\beta) = e^{-i\beta H_B}, \quad H_B = \sum_k X_k$$

Drives transitions between bitstrings — single-qubit only.

- Depth $p \;\Rightarrow\; 2p$ variational parameters.
- Larger p: more expressive, but harder to optimise and noisier.

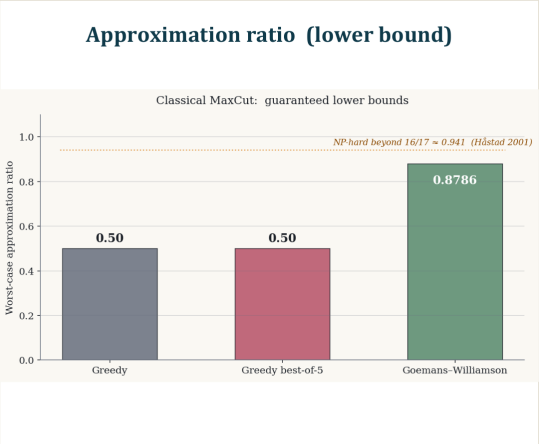
Classical benchmarks: greedy and Goemans–Williamson

Greedy. Process vertices in order; each joins the side cutting more already-assigned edges. Worst-case $r = 1/2$.

Greedy best-of-5. Run greedy 5 times with different orderings; keep the best cut.

Goemans–Williamson (1995). Relax to unit vectors, solve an SDP, round by a random hyperplane. Worst-case $r \geq 0.8786$, runtime $O(n^{3.5})$.

NP-hardness wall. Achieving $r > 16/17 \approx 0.941$ in polynomial time would imply $\mathrm{P} = \mathrm{NP}$ (Håstad 2001).

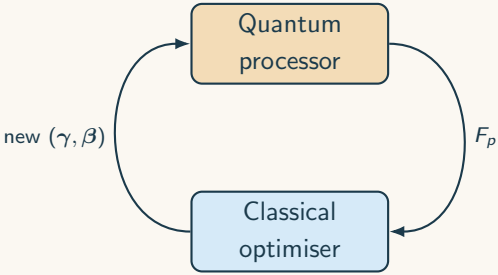


Implementation

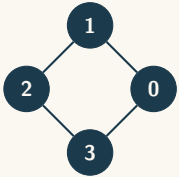
The hybrid loop

Each iteration:

1. Prepare $|\psi_p(\gamma, \beta)\rangle$ on the QPU.
2. Measure $F_p = \langle H_C \rangle$ via repeated sampling.
3. Send F_p to a classical optimiser (COBYLA, L-BFGS-B, SPSA ...).
4. Receive new (γ, β), repeat until converged.



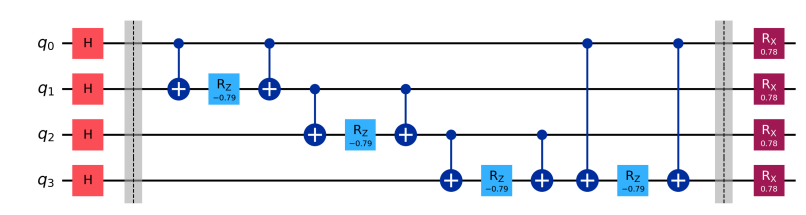
Explicit circuit: C_4, depth $p = 1$



C_4 : 4 vertices,

4 edges

$\gamma = 0.79, \ \beta = 0.39$



Reading left $\rightarrow$ right: **(1)** $H^{\otimes 4}$ to prepare $|+\rangle^{\otimes 4}$, **(2)** four cost blocks $U_C(\gamma)$, one per edge, **(3)** four single-qubit $R_X(2\beta)$ gates as the mixer $U_B(\beta)$.

How the circuit realises the ansatz

Cost unitary, edge by edge:

$$U_C(\gamma) = \prod_{(i,j)\in E} e^{-i\gamma(I - Z_i Z_j)/2}$$

Drop the global phase $e^{-i\gamma/2}$ per term:

$$e^{-i\gamma(I - Z_i Z_j)/2} \;\propto\; e^{+i\gamma\, Z_i Z_j/2}.$$

Standard identity:

$$e^{i\gamma Z_i Z_j/2} \;=\; \mathrm{CX}_{ij} \cdot R_Z(-\gamma)_j \cdot \mathrm{CX}_{ij}.$$

One CX–R_Z–CX block per edge.

Mixer unitary, qubit by qubit:

$$U_B(\beta) \;=\; e^{-i\beta \sum_k X_k} \;=\; \prod_k R_X(2\beta)_k.$$

One $R_X(2\beta)$ per qubit — no entangling gates.

Gate count for C_4, $p = 1$:

- $|E| = 4$ edges $\Rightarrow$ 4 cost blocks (8 CX, 4 R_Z).
- $n = 4$ qubits $\Rightarrow$ 4 R_X + 4 H.

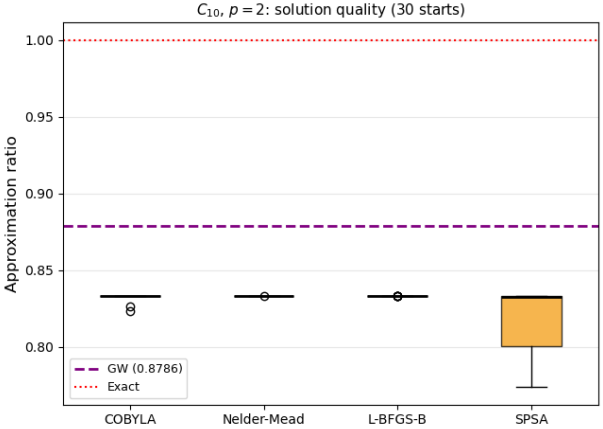
Choosing the classical optimiser

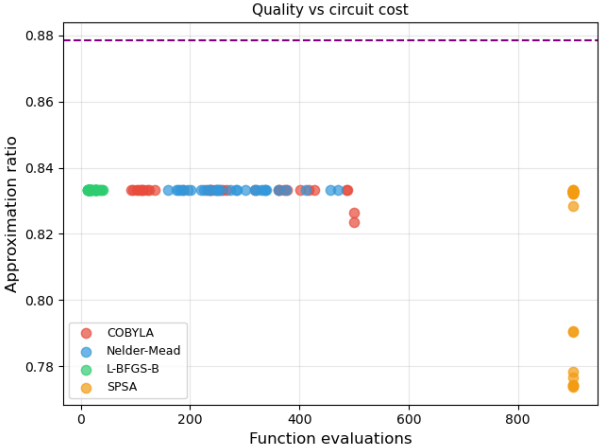
Optimiser	Gradient	Cost / step	Best for
COBYLA	none (trust-region)	$O(p)$	shot-noise robust
L-BFGS-B	parameter-shift	$2(\lvert E\rvert + n)p$	exact gradient (sim)
Nelder–Mead	none (simplex)	$O(p)$	fallback
SPSA	stochastic 2-eval	2	real hardware

Our choice: L-BFGS-B (primary) + COBYLA (cross-check)

L-BFGS-B converges fastest on noiseless statevector via the exact per-edge parameter-shift gradient. Running COBYLA in parallel as a gradient-free baseline rules out optimiser-specific artefacts. SPSA would replace both on real hardware.

Optimiser comparison



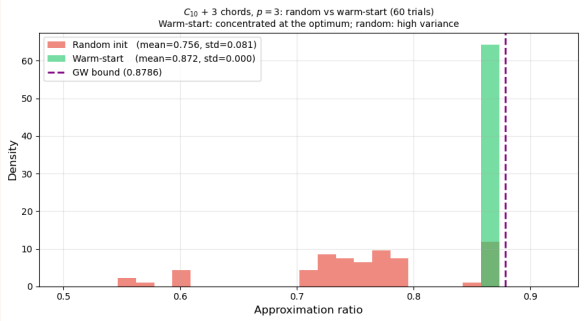


Warm start: initialise $p{+}1$ from the p-layer optimum

Algorithm (Zhou et al. 2020):

1. Optimise $p{=}1$ globally (random restarts).
2. For $k = 2, \ldots, p$:
   $\theta_0 \leftarrow (\boldsymbol{\gamma}^\star, \gamma^\star_{p-1}) \,\|\, (\boldsymbol{\beta}^\star, \beta^\star_{p-1}) + \eta$
   Local optimise F_k from θ_0.
3. Return final $(\boldsymbol{\gamma}^\star, \boldsymbol{\beta}^\star)$.

Intuition. A p-layer optimum is already a good region of parameter space for $p{+}1$ layers — the extra layer with small γ, β is nearly the identity.



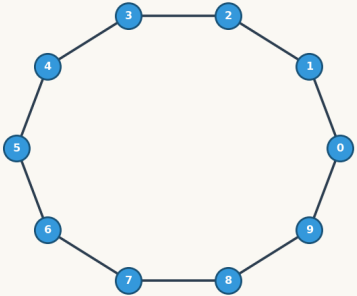
Warm-start collapses the variance: std drops from 0.083/0.126 to 0.000.

Results

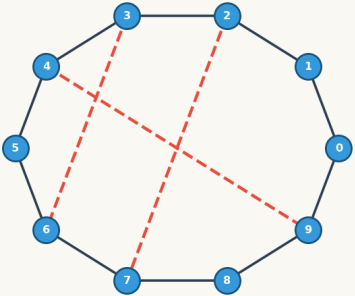
Problem instances

Graph Instances

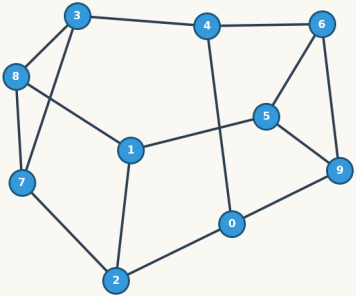
C_{10} **(cycle)**



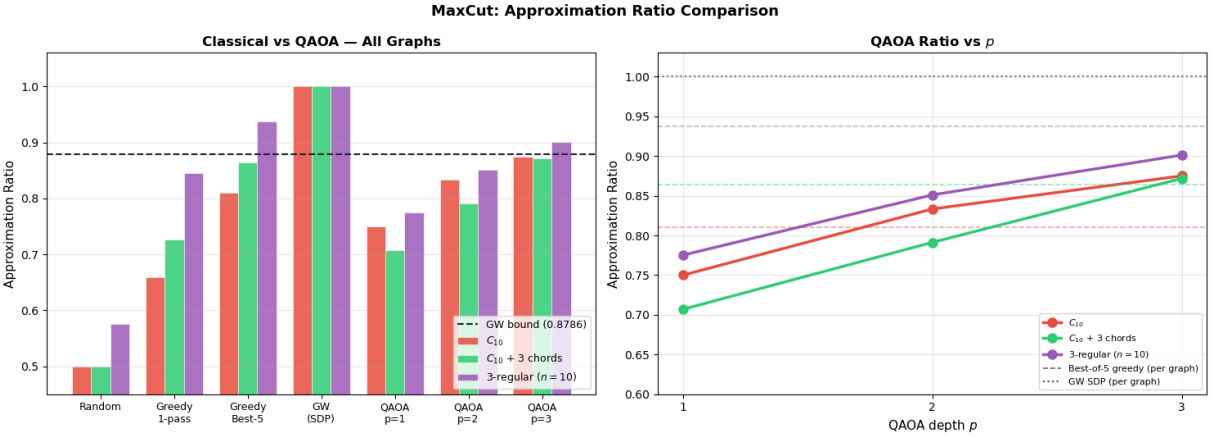
C_{10} **+ 3 chords**



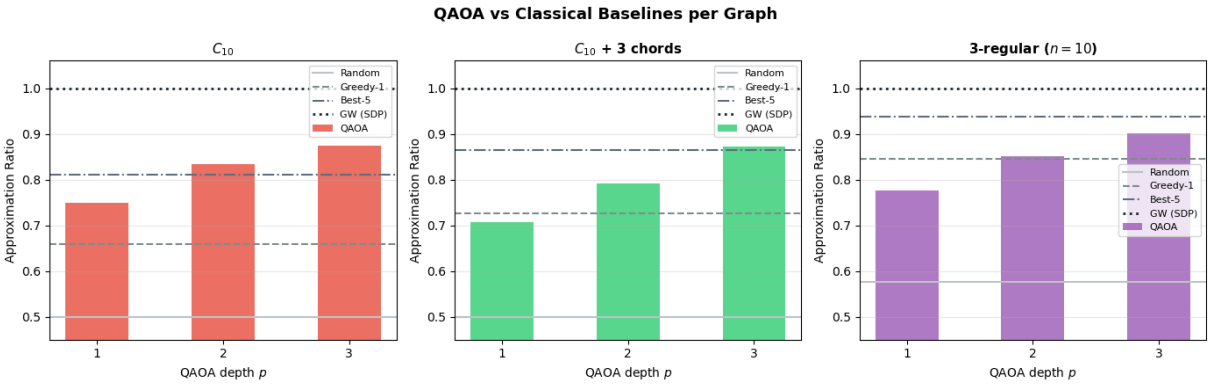
3-regular ($n = 10$)



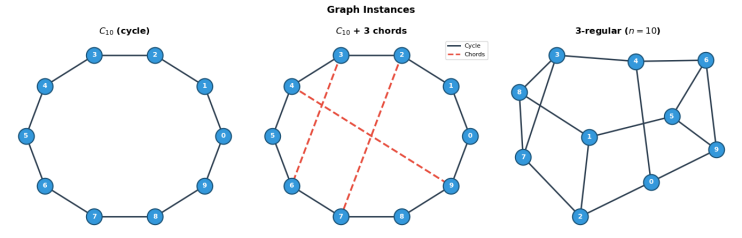
Performance comparison



Approximation ratio versus depth



Measurement distributions



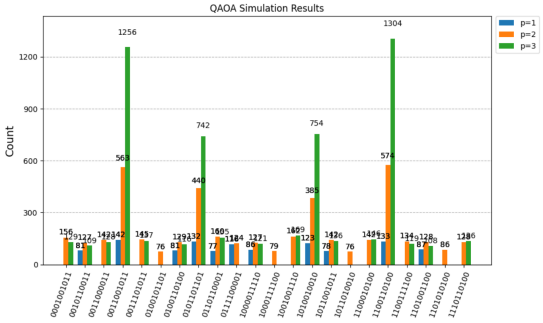
C_{10}:
$|1010101010\rangle$
$|0101010101\rangle$

C_{10} + **3 chords:**
same two states

3-regular:
$|1100110100\rangle$
$|0011001011\rangle$

Two solutions per graph because every cut has a $\mathbb{Z}_2$ complement-symmetry $z \mapsto \bar{z}$.

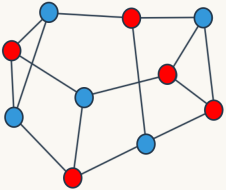
3-regular: not just one peak



4096 shots; $p = 1$ blue, $p = 2$ orange, $p = 3$ green.

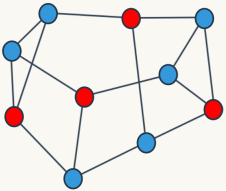
cut $= 13$ (MaxCut)

$|1100110100\rangle$, $|0011001011\rangle$



cut $= 12$ (2nd best)

$|0101101101\rangle$, $|1010010010\rangle$



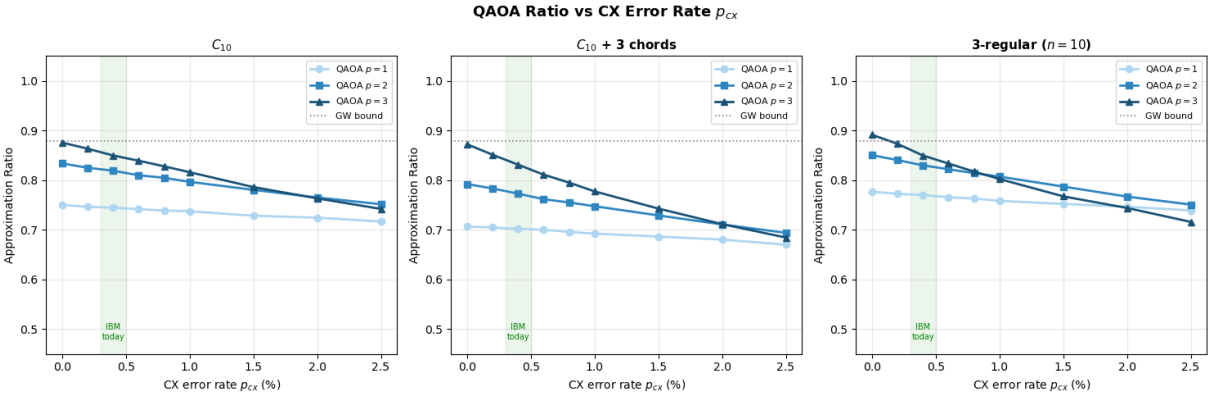
Introducing noise

After every CX gate

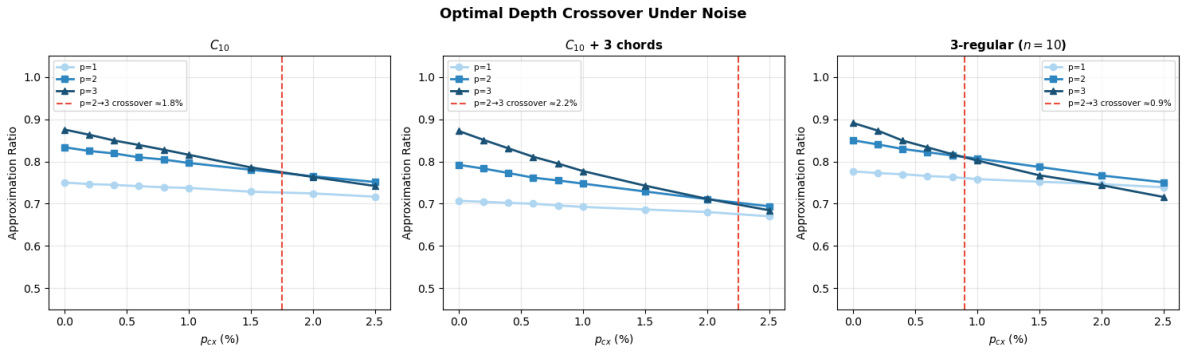
$$\mathcal{E}(\rho) \;=\; (1 - p_{cx})\,\rho \;+\; \frac{p_{cx}}{15} \sum_{P \in \{I,X,Y,Z\}^{\otimes 2} \setminus \{I^{\otimes 2}\}} P \rho P^{\dagger}$$

- Single-qubit gates (H, R_X, R_Z) carry the same form with rate $p_{cx}/10$ (1Q fidelities are typically $10\times$ better than 2Q).
- Total CX count $= 2p|E|$, so circuit fidelity decays as $(1 - p_{cx})^{2p|E|}$ — noise grows with depth and graph size.
- We sweep $p_{cx} \in [0, 0.025]$.

Ratio degradation with CX error rate



Crossover: when shallower beats deeper



- Total CX count $= 2p|E|$, so circuit fidelity decays as $(1 - p_{cx})^{2p|E|}$.
- Above $p_{cx} \approx 1\%$, $p{=}1$ **outperforms** $p{=}3$ on every graph tested.
- Reported median CX errors on superconducting hardware sit in the 0.5–1% range — close to the crossover region.

References i

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Thanks for your attention!

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Physics 565 / 656 · Spring 2026